FusionCore: A 23-State UKF for IMU, Encoder, GPS, and VSLAM Fusion in ROS 2: Quantitative Evaluation on Twelve Full-Length Sequences

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***Abstract*—FusionCore is an open-source ROS 2 sensor fusion package that fuses IMU, wheel encoder odometry, GPS, and Visual SLAM pose into a 100 Hz odometry stream using a 23-state Unscented Kalman Filter. The 23 states include gyroscope bias, accelerometer bias, and a wheel encoder yaw rate bias estimated online through GPS cross-covariance. We report results on twelve full-length NCLT sequences (55–92 min each, ∼940 min total) against `robot_localization` EKF and UKF. FusionCore achieves lower Absolute Trajectory Error on ten of twelve sequences, with improvements from $1.2\times$ to $22.2\times$. The `robot_localization` UKF diverges numerically on all twelve sequences.**

***Index Terms*—sensor fusion, unscented Kalman filter, ROS 2, GPS, IMU, state estimation**

I. Introduction and Motivation

Accurate pose estimation for ground robots requires fusing high-rate IMU data with low-rate GPS and wheel odometry. `robot_localization` [2] is the standard ROS solution, but several architectural choices limit practical performance: gyroscope and accelerometer biases are not filter states; GPS requires a separate `navsat_transform` node using UTM projection (subject to zone boundary failures); outlier rejection uses scalar Mahalanobis thresholds that do not account for measurement dimensionality; and noise covariance is fixed at configuration time.

FusionCore addresses each limitation in a single ROS 2 lifecycle node with the following contributions: (1) a 23-state UKF with continuous online estimation of gyroscope bias, accelerometer bias, and wheel encoder yaw rate bias; (2) ECEF-native GPS fusion eliminating UTM projection; (3) per-sensor chi-squared outlier gating calibrated per degrees of freedom; (4) automatic noise covariance adaptation from the innovation sequence; (5) ZUPT for stationary periods; (6) IMU ring-buffer retrodiction for GPS/VSLAM delay compensation; and (7) Visual SLAM pose fusion with automatic map-reinitialization recovery.

II. Filter Design

A. State Vector

The 23-dimensional state is: $\mathbf{x} = [\mathbf{p}, \mathbf{q}, \mathbf{v}, \boldsymbol{\omega}, \mathbf{a}, \mathbf{b}_g, \mathbf{b}_a, b_{\text{ewz}}]^\top$, where $\mathbf{p} \in \mathbb{R}^3$ is ENU position, $\mathbf{q}$ is unit quaternion, $\mathbf{v}, \boldsymbol{\omega}, \mathbf{a} \in \mathbb{R}^3$ are body-frame velocity, angular velocity, and acceleration, $\mathbf{b}_g, \mathbf{b}_a \in \mathbb{R}^3$ are gyroscope and accelerometer biases, and $b_{\text{ewz}} \in \mathbb{R}$ is the wheel encoder angular velocity Z bias.

The 23rd state is the key novelty. In differential drive robots, b_{ewz} arises from wheel radius mismatch and mechanical asymmetry. FusionCore identifies it online through GPS heading cross-covariance: when GPS is available, observed heading error back-propagates into b_{ewz}. During GPS blackouts (coast mode), the estimated bias is subtracted from encoder ω_z before integration, mirroring exactly how $\mathbf{b}_g$ handles gyroscope drift.

B. Key Design Choices

Chi-squared outlier gating. Every measurement is gated by $d^2 = \boldsymbol{\nu}^\top S^{-1} \boldsymbol{\nu} \leq \tau_s$, where τ_s is set per sensor per degrees of freedom: GPS 3-DOF $\tau = 16.27$ [$\chi^2(3, 0.999)$], VSLAM 6-DOF $\tau = 22.46$, heading 1-DOF $\tau = 10.83$. `robot_localization` exposes a single scalar threshold compared against $\sqrt{d^2}$, conflating dimensionality: calibrating for a 3-DOF sensor leaves a 1-DOF sensor under-protected.

Adaptive noise covariance. Sensor noise R adapts online: $R \leftarrow (1-\alpha)R + \alpha\hat{C}$, where $\hat{C}$ is the empirical innovation covariance from a 50-sample window and $\alpha = 0.01$. A floor $R_{ii} \geq R_{0,ii}$ prevents collapse. This is the primary mechanism through which FusionCore handles the GPS covariance mismatch described below.

III. Evaluation

A. Setup

We evaluate on the NCLT dataset [1]: Microstrain 3DM-GX3-45 IMU (100 Hz), Segway wheel odometry (100 Hz), Novatel SPAN-CPT GPS (5 Hz). All twelve available sequences are run to full length (55–92 min, ∼940 min total). Metric is ATE RMSE (meters) computed with EVO [3] against RTK ground truth. FusionCore uses a single `nclt_fusioncore.yaml` across all twelve sequences with no per-sequence tuning. `robot_localization` EKF is configured with chi-squared-equivalent thresholds: `odom0_twist_rejection_threshold: 4.03` ($\sqrt{\chi^2(3, 0.999)}$) and `odom1_pose_rejection_threshold: 3.72`.

TABLE I
ATE RMSE (METERS) ON TWELVE FULL-LENGTH NCLT SEQUENCES.

Sequence	FC	RL-EKF	RL-UKF	Winner
2012-01-08	**18.6**	41.2	†	FC (2.2×)
2012-02-04	**49.7**	265.5	†	FC (5.3×)
2012-03-31	**22.0**	156.5	†	FC (7.1×)
2012-05-11	**9.7**	11.5	†	FC (1.2×)
2012-06-15	49.2	**18.2**	†	RL (2.7×)
2012-08-20	98.3	**10.6**	†	RL (9.3×)
2012-09-28	**10.8**	55.7	†	FC (5.2×)
2012-10-28	**29.9**	60.0	†	FC (2.0×)
2012-11-04	**60.1**	122.0	†	FC (2.0×)
2012-12-01	**21.0**	90.7	†	FC (4.3×)
2013-02-23	**59.4**	82.2	†	FC (1.4×)
2013-04-05	**12.1**	268.9	†	FC (22.2×)

† Numerical divergence (NaN) within first 30 s on all sequences.

B. Results

C. The GPS Covariance Mismatch Problem

The NCLT NavSatFix stream reports `var_xy = 9` (3 m sigma), matching the Novatel SPAN-CPT open-sky specification. Measured against RTK ground truth, the actual p95 error ranges from 9.7 m to 53.1 m across sequences: 2 to 18 times the stated sigma. RL-EKF trusts the stated covariance and calibrates its chi-squared gate accordingly. When actual errors reach 40–200 m, those fixes land far outside the gate and get rejected. With GPS effectively disabled, RL-EKF reverts to wheel-encoder dead-reckoning, producing drift rates of 31.84 m/km and 50.11 m/km on the two worst sequences, diagnostic of open-loop operation across full runs.

FusionCore's adaptive noise window inflates the GPS noise model in real time from the innovation sequence, recalibrating the chi-squared gate to actual error levels. When GPS actually is bad (2012-08-20 adversarial cluster, analyzed below), the gate still rejects it. The filter no longer permanently discards a sensor whose manufacturer-reported sigma is optimistic.

D. Analysis of the Two FusionCore Losses

2012-06-15 (FC 49.2 m, RL 18.2 m). The dataset's GPS-sparsest sequence: 12,399 mode-3 fixes vs. 17,000–22,000 on all others, with a 462-second (7.7-min) blackout. During coast mode, FusionCore dead-reckons on encoder and IMU with b_{ewz} subtracted. Any residual b_{ewz} error compounds over 7.7 min: at 100 Hz, even a small heading rate residual accumulates quadratic lateral error. RL-EKF wins through a structural advantage: its two-dimensional mode has fewer divergence degrees of freedom, not because of better fusion. The architecturally correct fix is magnetometer integration providing absolute heading during GPS absence.

2012-08-20 (FC 98.3 m, RL 10.6 m). The GPS stream contains 105 mode-3 fixes located 720–840 m from RTK ground truth, concentrated in a 24-second window at the end of a 211-second blackout. These appear valid in the raw data stream. Coast mode slightly relaxes the chi-squared gate to re-acquire GPS after blackout; the tight cluster of adversarial fixes each individually pass the gate under elevated position uncertainty and collectively pull the estimate. RL-EKF rejects them because its miscalibrated gate (which causes 10 other sequence losses) incidentally also rejects these. The fix is a velocity-consistency pre-check upstream of chi-squared gating: a fix 720 m from the dead-reckoned position after 211 s implies 3400 m/s travel speed. A `max_implied_speed` check (e.g., 20 m/s) handles this with zero cost to normal operation.

IV. Conclusion

FusionCore is a 23-state ROS 2 UKF fusing IMU, wheel encoders, GPS, and VSLAM at 100 Hz in a single lifecycle node. On twelve full-length NCLT sequences, it outperforms `robot_localization` EKF on 10 of 12, with gains from 1.2× to 22.2×. The RL UKF diverges numerically on all twelve. The two FC losses are attributable to diagnosable causes with identified fixes. A full system description and evaluation are available in the companion technical report [4]. FusionCore is available at https://github.com/manankharwar/fusioncore (Apache 2.0).

References

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